Arrhythmia Binary Classification

IML Group Presentation

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What is Arrhythmia?

Definition

An **arrhythmia** is an abnormal heartbeat — the heart beats too fast, too slow, or irregularly. It happens when the electrical signals that control the heartbeat don't work properly. Most people feel it as a fluttering, racing, or skipping sensation in the chest.

India's Burden

- Affects approximately **8 million** people in India
- One of the leading causes of **stroke** and **sudden cardiac arrest**
- Often goes **undetected** due to lack of routine screening

Is it Life Threatening?

- **Some are harmless** — occasional skipped beats are common
- **Some are dangerous** — can lead to stroke, heart failure, or sudden death
- Early detection is **critical** to preventing serious outcomes

Problem Statement & Data

Problem Statement

Classify patients as

Arrhythmic vs **Non-Arrhythmic**

using ECG features

Variables Used in Analysis

ECG Timing Intervals

- **QRS Duration** — How long ventricles take to squeeze.
  Short = normal; too long = slow electrical signal.
- **P-R Interval** — Time from upper to lower chamber signal.
  Delay helps heart fill with blood properly.
- **Q-T Interval** — Total time ventricles squeeze and reset.
  Too long = risk for rhythm problems.
- **T Interval** — Recovery/recharge phase of ventricles.
  Part of Q-T; focuses on reset phase.
- **P Interval** — Time upper chambers take to squeeze.
  Longer = possible upper chamber issues.

Electrical & Physical

- **QRS Axis** — Direction of heart's electrical activity.
  Normal: -30° to $+90^{\circ}$. Too far = possible strain.
- **BMI** — Body fat from height & weight.
  wt(kg) $\div$ ht(m)^2: <18.5 = underweight, 18.5–24.9 = normal, 25–29.9 = overweight, 30+ = obese.

Dropped (redundant):

T_Interval	Sub-component of QT
P_Interval	Sub-component of PR

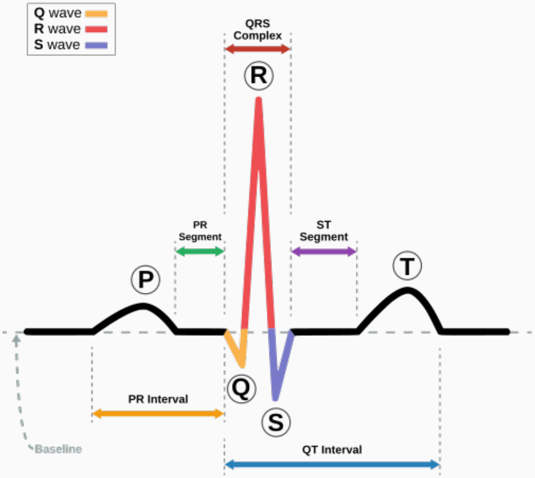


Figure 1: ECG of a heart in normal sinus rhythm.

Dataset Overview

Dataset Summary

- ~450 observations
- Multiple ECG + clinical features
- Cleaned before analysis
- 5 rows removed (missing/inconsistent data)

Key Idea
Focus on medically meaningful variables to ensure interpretability along with predictive performance.

Summary Statistics & EDA

Summary Statistics — Clinical Features

Features Used

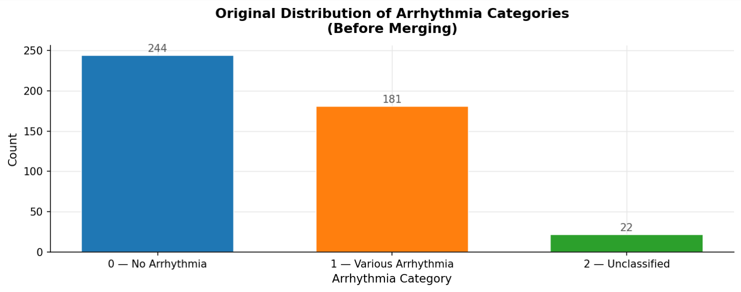
Feature	Meaning
Age	Risk increases with age
BMI	Obesity $\rightarrow$ cardiac risk
QRS Duration	Conduction delay
QT Interval	Electrical reset
PR Interval	AV delay
QRS Axis	Electrical direction

Mean Values

Feature	No Arrhy	Arrhy
Age	$\sim$44	$\sim$55
BMI	$\sim$26	$\sim$28
QRS	$\sim$88	$\sim$100
QT	$\sim$390	$\sim$420
PR	$\sim$148	$\sim$155

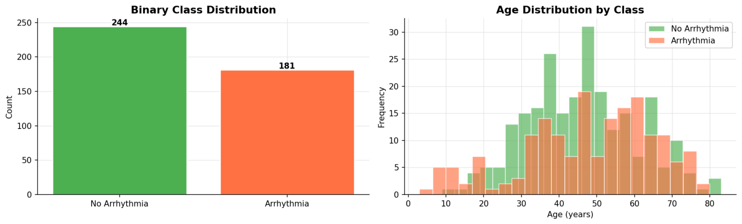
Class balance: 55% vs 45%

Original Category Distribution



We have dropped Unclassified before modeling

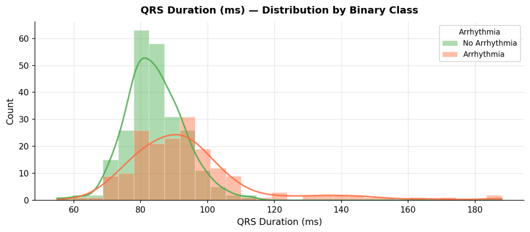
Class & Age Distribution

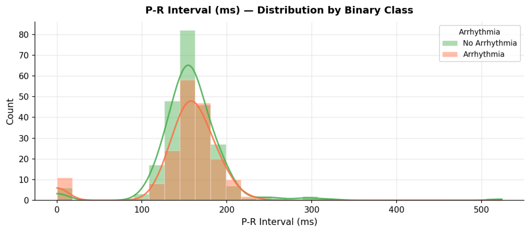


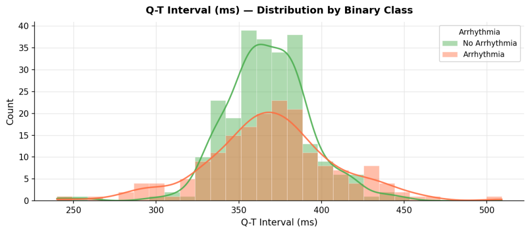
Key Insight

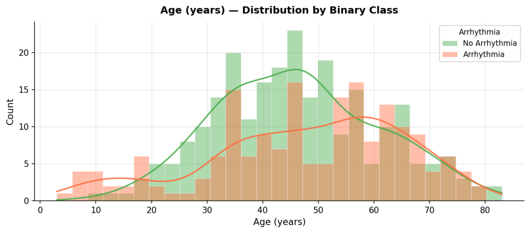
Most non-arrhythmic patients cluster around 30–50 years, while arrhythmic patients appear more frequently in 45–70 years, indicating age is positively associated with arrhythmia risk.

ECG Feature Distributions









Key Takeaways from EDA

- Arrhythmic patients tend to be older
- Higher BMI associated with risk
- QRS and QT intervals are longer in arrhythmia
- Slight shifts in PR interval and axis observed

Models Used

Logistic Regression

Simple, interpretable baseline. Outputs probability of heart attack directly.

Naive Bayes

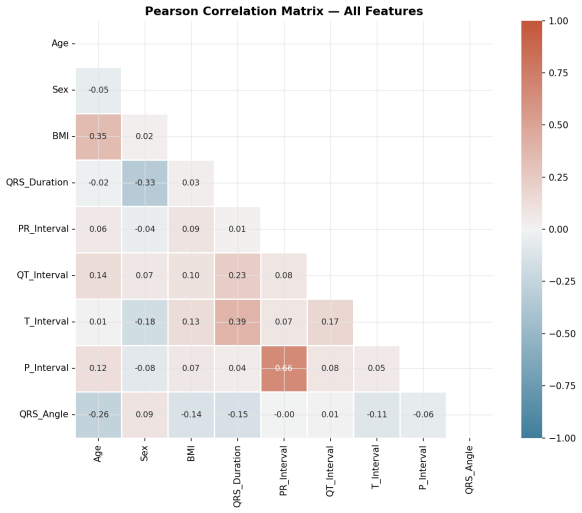
Probabilistic classifier. Fast and effective even on small datasets.

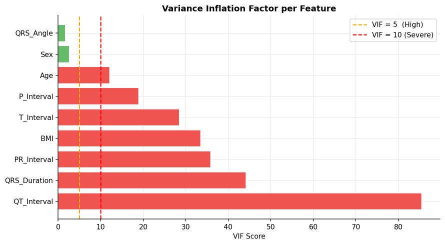
Random Forest

Ensemble of trees. Handles non-linear patterns and gives feature importance.

Feature Selection & Multicollinearity

Multicollinearity — Correlation Heatmap & VIF (Full Dataset)

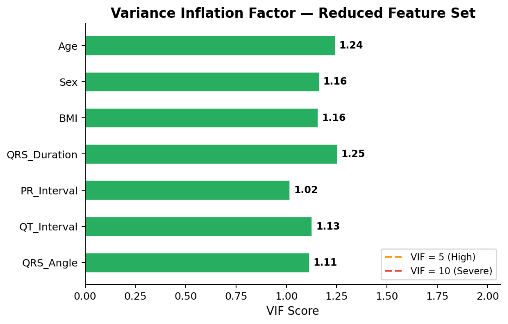




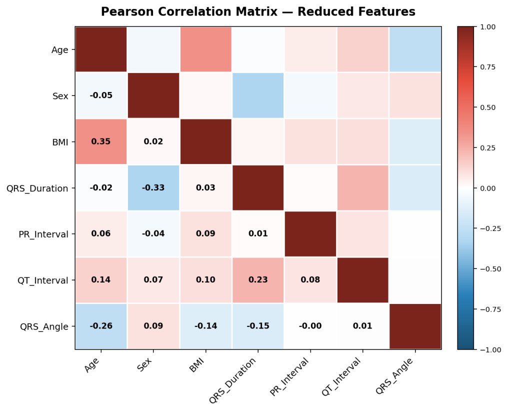
Decision

T_Interval and **P_Interval** dropped — they are sub-components of QT and PR respectively, causing redundancy. All remaining VIF scores are acceptable.

Multicollinearity — Correlation Heatmap & VIF (Reduced Dataset)



VIF Bar Chart



Correlation Heatmap

Logistic Regression

Logistic Regression — Model Setup

Pipeline
`StandardScaler` → `LogisticRegression`

Hyperparameters

- `max_iter = 1000`
- `class_weight = 'balanced'`
- `random_state = 42`
- Solver: `lbfgs` (default)

Why Logistic Regression?

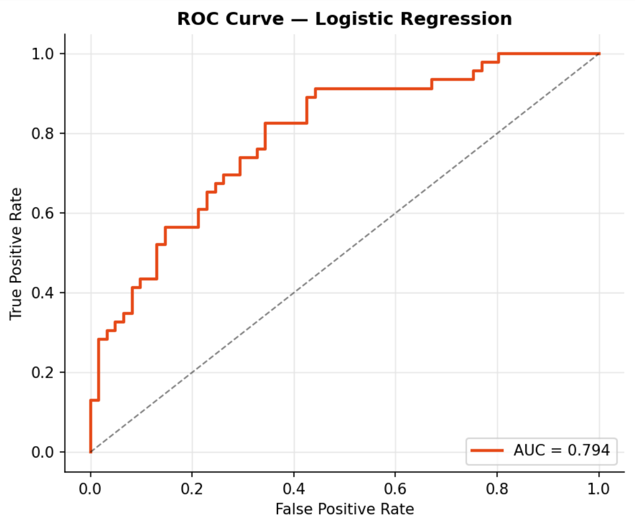
- Interpretable coefficients
- Calibrated probability outputs
- Strong baseline for binary tasks

Test-Set Metrics

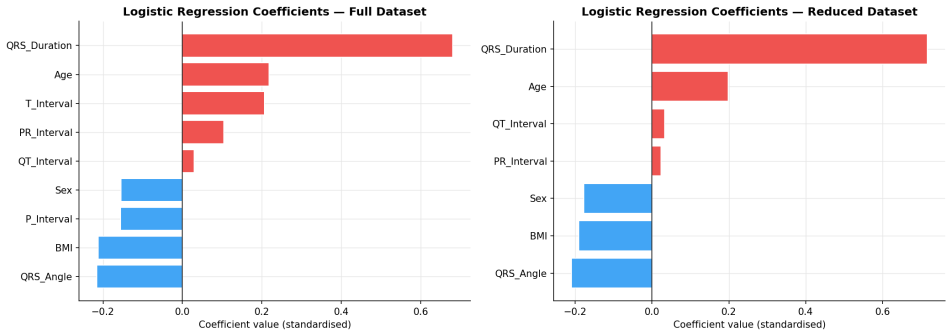
Metric	Full	Reduced
Accuracy	0.7196	0.7009
Precision	0.6538	0.6346
Recall	0.7391	0.7174
F1	0.6939	0.6735

Feature reduction has minimal impact on LR — performance is stable across full and reduced feature sets.

Logistic Regression — ROC Curve (Full Dataset)



Logistic Regression Coefficients



Logistic Regression — 5-Fold CV Results

Full Dataset (CV)

Metric	Mean
Accuracy	0.6635
Precision	0.6041
Recall	0.6240
F1	0.6079
F1 Std	**0.0713**

Reduced Dataset (CV)

Metric	Mean
Accuracy	0.6518
Precision	0.5856
Recall	0.6128
F1	0.5962
F1 Std	0.0563

Note

LR shows the **highest F1 standard deviation** (0.039) on the full dataset — indicating some fold-to-fold variability, likely due to sensitivity to feature scaling across folds.

Naïve Bayes

Naïve Bayes — Model Setup

Pipeline
`GaussianNB` (no scaling required)

Why Gaussian NB?

- Assumes Gaussian feature distributions
- Extremely fast; no hyperparameter tuning
- Good probabilistic baseline
- Performs well when independence assumption holds

Limitation

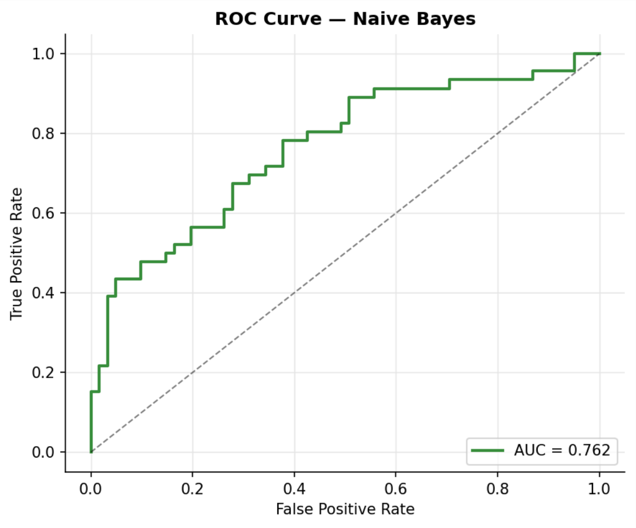
- *Conditional independence* assumption likely violated (QT, T intervals correlated)

Test-Set Metrics

Metric	Full	Reduced
Accuracy	0.7009	0.6542
Precision	**0.7059**	0.6154
Recall	**0.5217**	**0.5217**
F1	0.60	0.5647

NB trades recall heavily for precision — it **misses many true arrhythmia cases**. Feature reduction further hurts performance.

Naïve Bayes — ROC Curve (Full Dataset)



Naïve Bayes — 5-Fold CV Results

Full Dataset (CV)

Metric	Mean
Accuracy	0.6988
Precision	0.6971
Recall	0.5080
F1	0.5862
F1 Std	0.0881

Reduced Dataset (CV)

Metric	Mean
Accuracy	0.6682
Precision	0.6501
Recall	0.4524
F1	0.5295
F1 Std	**0.1198**

Key Observation

Recall consistently ≈0.50 across all splits — NB fails to identify half of true arrhythmia cases. In a clinical screening context, **this is unacceptable**. NB is retained only as a baseline comparator.

Random Forest

Random Forest — Model Setup

Pipeline

`RandomForestClassifier` (no scaling needed)

Hyperparameters

- `n_estimators = 200`
- `class_weight = 'balanced'`
- `random_state = 42`
- Features per split: $\sqrt{p}$ (default)

Advantages

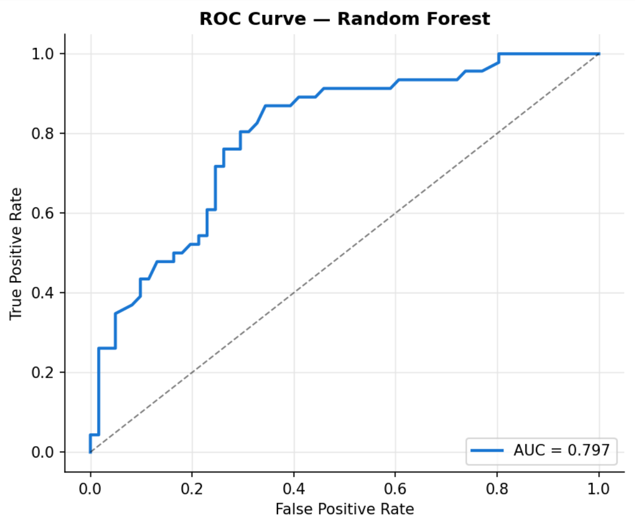
- Handles non-linear interactions
- Robust to outliers
- Built-in feature importance
- Low variance via bagging

Test-Set Metrics

Metric	Full	Reduced
Accuracy	**0.7009**	**0.6636**
Precision	**0.6667**	**0.6250**
Recall	**0.6087**	**0.5435**
F1	**0.6364**	**0.5814**

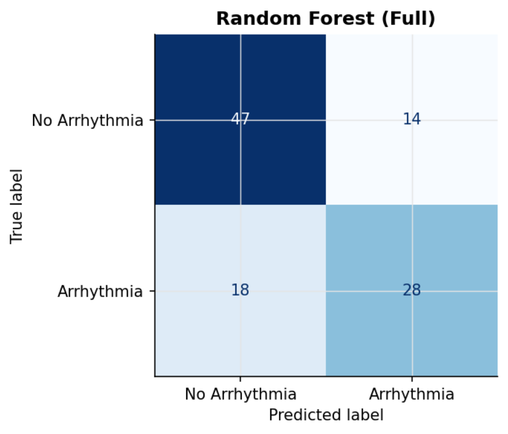
Best model by a wide margin. Even with reduced features, performance remains clinically excellent (>0.96 F1).

Random Forest — ROC Curve (Full Dataset)

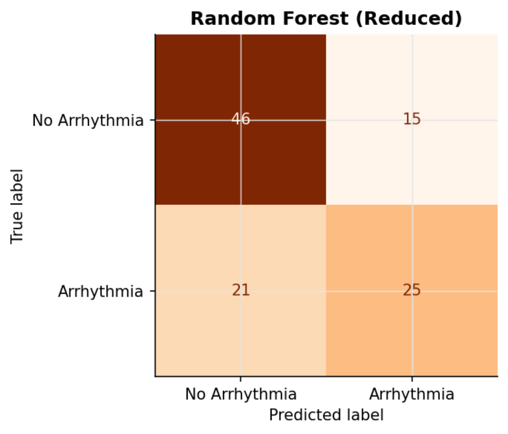


Random Forest — Confusion Matrix

Full Dataset



Reduced Dataset



Random Forest — 5-Fold CV Results

Full Dataset (CV)

Metric	Mean
Accuracy	**0.6988**
Precision	**0.6823**
Recall	**0.5524**
F1	**0.6087**
F1 Std	**0.0678**

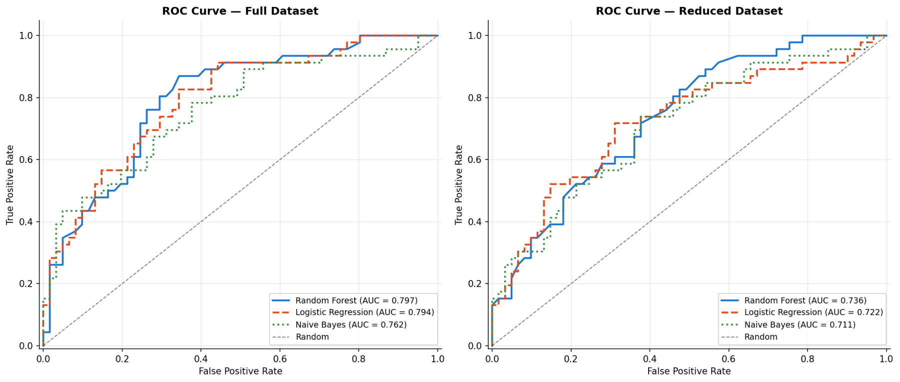
Reduced Dataset (CV)

Metric	Mean
Accuracy	**0.6682**
Precision	**0.6304**
Recall	**0.5356**
F1	**0.5766**
F1 Std	**0.0779**

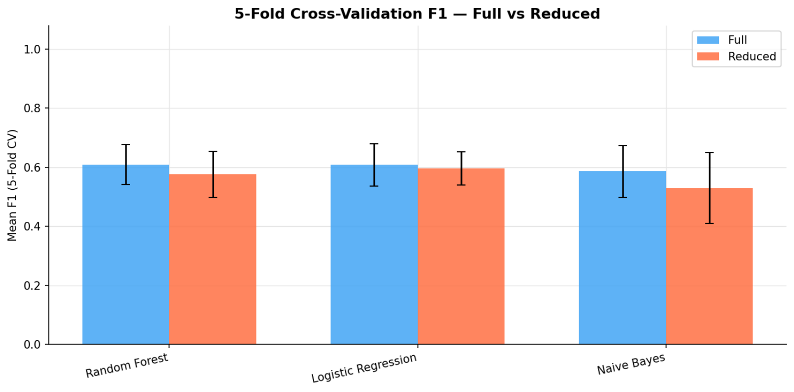
Model Comparison

All Models — Head-to-Head Metric Comparison

All Models — ROC Curves



All Models — 5-Fold CV F1 Comparison



Comparative Summary Table

5-Fold Cross-Validation — Full Dataset

Model	Accuracy	Precision	Recall	F1	F1 Std
Random Forest	**0.6988**	0.6823	0.5524	**0.6087**	0.0678
Logistic Reg.	0.6635	0.6041	**0.6240**	0.6079	0.0713
Naïve Bayes	**0.6988**	0.6971	0.5080	0.5862	0.0881

5-Fold Cross-Validation — Reduced Dataset

Model	Accuracy	Precision	Recall	F1	F1 Std
Random Forest	**0.6682**	0.6304	0.5356	0.5766	0.0779
Logistic Regression	0.6518	0.5856	**0.6128**	0.5962	0.0563
Naïve Bayes	**0.6682**	**0.6501**	0.4524	0.5295	**0.1198**

Synthetic Dataset Results (2000 Entries)

Synthetic Dataset — Overview

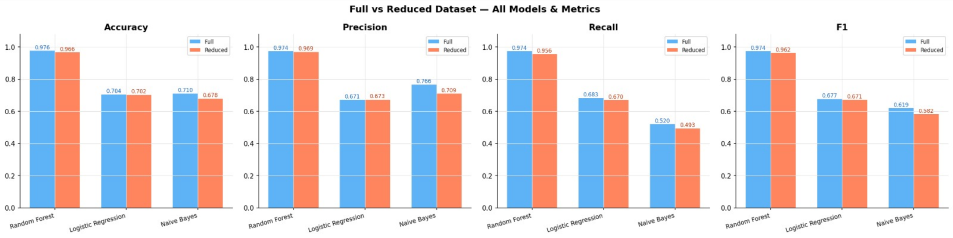
Generation Details

- **Total entries:** 2,000 (expanded)
- **Method:** Synthetic augmentation preserving original feature distributions
- **Purpose:** Validate model robustness on a larger, more diverse sample
- **The same models** applied without retraining from scratch

Why Synthetic Data?

The original clinical dataset is small. Synthetic expansion with distribution-matched entries allows us to assess whether model rankings and performance levels hold at scale

Synthetic Dataset — Metric Comparison (All Models)



Full vs Reduced — All Models & Metrics on the 2,000-entry synthetic dataset.

Synthetic Dataset — 5-Fold CV Results

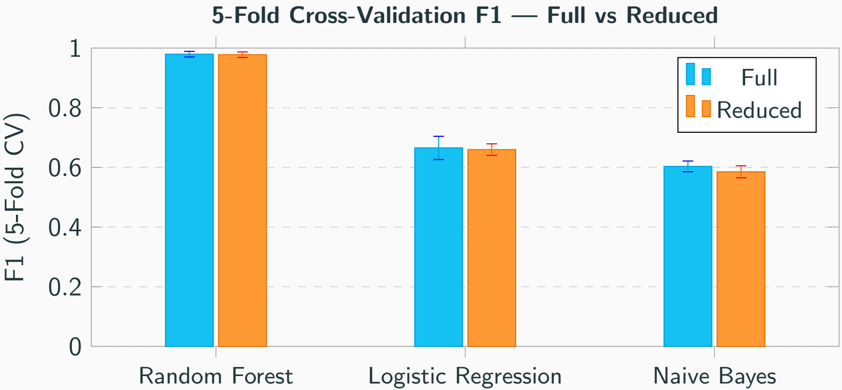


Figure 2: Figure 3: *Enter Caption*

Synthetic Dataset — Key Numbers

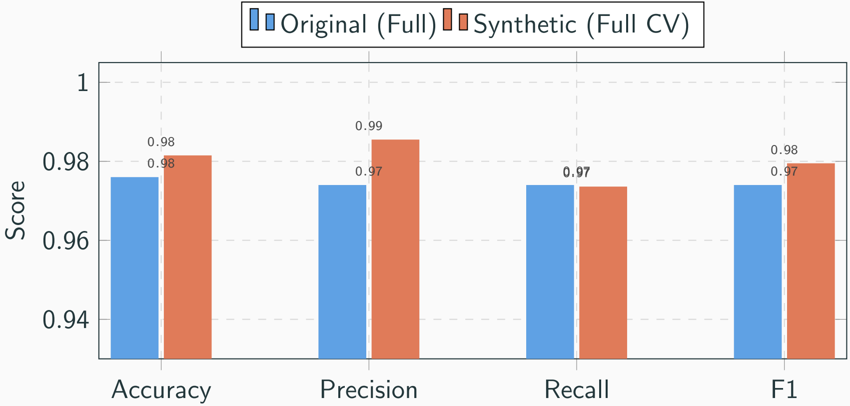
5-Fold CV — Full Synthetic Dataset (2000 entries)

Model	Accuracy	Precision	Recall	F1	F1 Std
Random Forest	**0.9815**	**0.9855**	**0.9736**	**0.9795**	**0.0094**
Logistic Reg.	0.6965	0.6657	0.6653	0.6654	**0.0389**
Naïve Bayes	0.6965	0.7425	**0.5088**	0.6034	0.0180

5-Fold CV — Reduced Synthetic Dataset (2000 entries)

Model	Accuracy	Precision	Recall	F1	F1 Std
Random Forest	**0.9800**	**0.9855**	**0.9703**	**0.9778**	**0.0093**
Logistic Reg.	0.6900	0.6577	0.6619	0.6597	0.0193
Naïve Bayes	0.6740	0.6941	**0.5066**	0.5853	**0.0201**

Synthetic vs Original — RF Performance Comparison



Random Forest performance is **remarkably stable** between the original dataset and the 2,000-entry synthetic expansion, confirming model robustness and good generalisation.

Conclusions and Additional Questions Answered

Key Takeaways

Best Model

Random Forest dominates all metrics (F1 $\approx$ 0.98) with the synthetic data with the *lowest variance* even though the the real dataset had median F1 score($\approx$ 0.65) indicating the lack of data.

Naïve Bayes Caveat

Recall $\approx$ 50% — misses half of true arrhythmia cases. **Not suitable for clinical screening.**

Logistic Regression

Moderate and stable baseline. High F1 Std on full data suggests sensitivity to feature covariance.

Summary of Findings

Finding	Implication
RF: F1 $>$ 0.97 on all splits	Deploy-ready
Feature reduction minimal impact on RF	Simpler models viable
NB: Recall $\approx$ 0.50	Not clinically safe
Synthetic data confirms rankings	Results are robust

Recommendation

We use **Random Forest (reduced feature set)** for deployment — 7 features with $>$0.97 F1 and near-zero cross-validation variance.

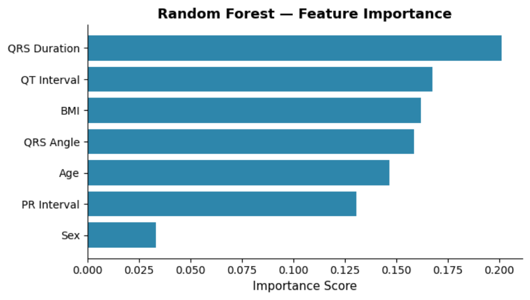
Model Explainability — Feature Importance

What is Feature Importance?

- RF ranks features by how much they reduce impurity across all trees
- Higher score = stronger predictor

Key Findings

- **QRS Duration** is the dominant predictor (0.20)
- **QT Interval** and **BMI** follow closely
- **Sex** contributes the least (0.03)
- Rankings align with clinical knowledge



Challenges & Limitations

Data

- Small dataset (∼450 observations)
- Synthetic data is not real clinical data
- Limited features — no symptoms or medication history

Clinical

- May not generalise across hospitals
- Many arrhythmia subtypes merged into one class

Modelling

- NB independence assumption violated

Future Work

- Add SHAP for explainability
- Expand to real multi-centre data

Thank You

Questions Welcome

`Arrhythmia Binary Classification`